PRISM: A General Reconstruction–Refinement Framework for Time Series Imputation

Anonymous CVPR submission

Abstract

Time-series imputation supports reliable analysis in energy, environmental monitoring, finance, healthcare and wearable sensing. Existing recurrent, attention-based and diffusion models can reconstruct overall trajectories, yet their imputed values may still deviate substantially from the true values when observations are sparse or consecutive entries are missing. To address this problem, we introduce PRISM, a general reconstruction–refinement framework that unifies point–patch evidence reconstruction with observation-guided residual correction. Specifically, PRISM comprises four components: (1) Point-Level Context Encoding (PCE), (2) Multi-Source Patch Evidence Construction (MPEC), (3) Pointwise Evidence-Guided Reconstruction (PEGR), and (4) Observation-Guided Residual Refinement (OGRR). PCE extracts temporal and cross-variable context, MPEC organizes temporal, cross-variable and training-history patches, and PEGR integrates point-level context with patch evidence to form an initial reconstruction. OGRR subsequently combines a temporal-context-driven neural residual with an observation-reference-guided gated correction, improving recovery at missing positions while preserving visible measurements. Experiments on six datasets across diverse domains and four missingness configurations demonstrate consistently competitive performance, with best or second-best results in multiple settings. OGRR also improves other imputation backbones while keeping their weights frozen and demonstrates strong standalone recovery without a preceding learned reconstruction network. Our work establishes the value of combining point–patch evidence reconstruction with observation-guided correction, offering a path toward reusable time-series recovery models across domains.

1. Introduction

Multivariate time-series analysis is important across energy, meteorology, finance, air quality, healthcare and wearable sensing. Missing data are a central concern because they affect downstream tasks such as forecasting and classification (Wang et al., 2025; Wu et al., 2023). For example, clinical records contain measurements collected at unequal intervals (Che et al., 2018; Silva et al., 2012), while wearable activity data combine signals from sensors attached to different body locations (Reiss and Stricker, 2012). Sensor failures, communication losses and irregular acquisition leave these records incomplete (Cini et al., 2022; Liu et al., 2023). Missing entries can be isolated or form consecutive gaps, with their locations varying across variables. These variations change which observations are available to each missing point, making accurate recovery more challenging.

Temporal continuity provides a direct starting point for imputation because observed neighboring states constrain how a signal changes through a gap. Early work on irregular clinical records incorporates observation masks and time intervals into recurrent modeling (Che et al., 2018); bidirectional temporal dependencies subsequently enable joint learning of sequence representations and missing values, with evaluation on clinical, air-quality and human-activity data (Cao et al., 2018). As consecutive entries disappear, however, the distance between a target and actual observations increases, weakening the direct support for estimates propagated through time alone. This motivates drawing complementary information from related variables, with graph message passing providing one mechanism for exchanging evidence between channels in traffic, air-quality and smart-grid imputation (Cini et al., 2022). Attention further broadens access across time and variables. Masking constraints support reconstruction from nonadjacent observations in clinical, air-quality and electricity data (Du et al., 2023), while low-rank structure extends spatiotemporal extraction to traffic networks, solar generation and energy consumption (Nie et al., 2024). Source variables may themselves be incomplete, and their corrupted features can affect information transfer, making reliability assessment important even when more context is available. T1 addresses this issue by binding temporal convolutional channels to variable-attention heads, controlling cross-variable information flow for different temporal features and evaluating it on energy, weather, healthcare and exchange-rate data, among other domains (Park et al., 2026). This progression shows that imputation requires both access to contextual evidence and a way to judge its usefulness under the current observa-

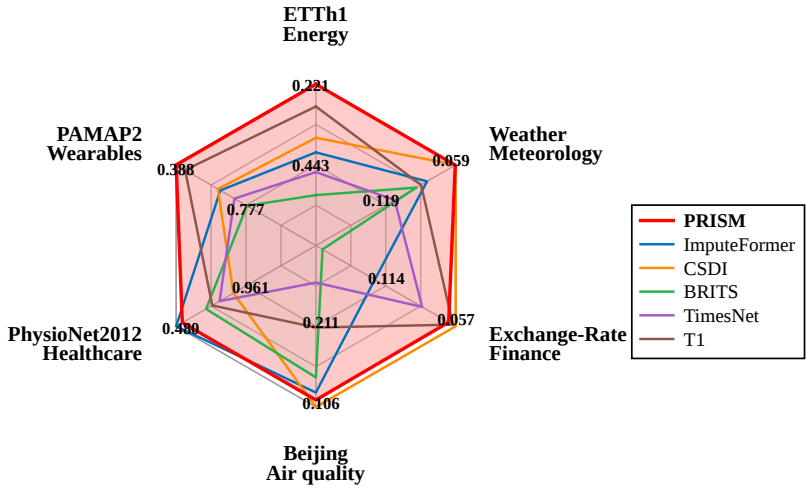


Figure 1. Cross-domain imputation performance of PRISM and baseline models over six datasets. Each radar axis reports the mean full-window MSE across four missingness configurations. Radius is the best mean MSE divided by the model's mean MSE within that dataset; larger radius indicates lower error, and axis ticks show the corresponding MSE values.

tion pattern.

The granularity of contextual representations further determines what information a model can extract. A time point corresponds directly to the value being recovered, while a local fragment describes a rise, decline or transition. Encoding temporal patches preserves such changes as a unit (Nie et al., 2023); organizing representations around variables supports relationships between distinct measurement processes (Liu et al., 2024a). Periodic and multi-scale representations further show that useful patterns can occur over different temporal ranges (Wang et al., 2025; Wu et al., 2023). For imputation, these advances suggest that a model should retain both the target point's local state and the shape of fragments that explain its variation, then select and combine this information according to the missing position.

Beyond representing evidence, a further question is how to turn it into accurate numerical estimates. Conditional diffusion learns missing-value distributions given visible observations to recover clinical physiological records and air-quality measurements (Tashiro et al., 2021); spatiotemporal priors further strengthen this conditioning for air-quality and traffic-speed imputation (Liu et al., 2023). Pursuing better numerical estimates, RDPI first generates an initial result and then models its residuals with conditional diffusion (Liu et al., 2024b). These developments motivate explicitly modeling correction after an initial estimate. Within an actual gap, a reconstructed trajectory can follow the overall trend while deviating in local level or transition amplitude. Remaining observations supply anchors for these quantities, motivating a connection between trajectory reconstruction and observation-guided pointwise correction.

This distinction motivates two connected design requirements. First, the useful evidence for a missing point depends on both its context and its position within a gap. A same-variable patch can reveal a continuation, another variable can expose a shared transition, and a recurring training-history fragment can supply a relevant local pattern. A point near an observed boundary and a point deep inside a gap may require different combinations of these fragments. Point-level context provides a target-specific description, whereas a patch preserves the shape of a local trajectory. We therefore organize explicit patch evidence around the target and let individual points determine how to read it, combining point and patch information within the same reconstruction process.

Second, the remaining observations retain numerical information after an initial trajectory has been reconstructed. Two estimates with similar shapes can differ in their level, transition amplitude or values near observed boundaries. Temporal features can identify a systematic reconstruction error, while references derived from actual visible values provide anchors for correcting it. Combining the initial estimate with visible measurements allows temporal context to inform residual prediction and observation-derived references to guide numerical correction at missing positions. This process connects local trajectory reconstruction with pointwise correction to further improve the initial recovery.

To this end, we introduce PRISM, a unified reconstruction–refinement framework comprising (1) Point-Level Context Encoding (PCE), (2) Multi-Source Patch Evidence Construction (MPEC), (3) Pointwise Evidence-Guided Reconstruction (PEGR), and (4) Observation-Guided Residual Refinement (OGRR). PCE combines residual temporal convolution with cross-

variable attention to represent each target. MPEC organizes same-variable temporal patches, cross-variable patches and eligible training-history members into a common evidence bank. PEGR reads this bank separately for each target point and jointly uses the query and retrieved evidence to correct a contextual base estimate. Confidence-weighted overlap fusion then forms the initial reconstruction. OGRR revisits this reconstruction alongside the visible observations, combining a temporal-context-driven neural residual with an observation-reference-guided gated correction. The reconstruction model stays frozen during OGRR training, and visible measurements are preserved in the final output.

We evaluate PRISM on six datasets covering energy, meteorology, finance, air quality, healthcare and wearable sensing, under four configurations that vary missingness density and gap length. Figure 1 summarizes the cross-domain results. PRISM delivers consistently competitive recovery and achieves best or second-best performance in multiple settings. Paired component studies examine patch evidence before and after refinement, while shared-refinement experiments show that OGRR also improves other frozen imputation backbones. Experiments without a preceding learned reconstruction network further reveal OGRR's standalone recovery capability. Our contributions are summarized as follows:

- We introduce PRISM, a unified framework that couples point–patch evidence reconstruction with observation-guided residual refinement. Source-organized patch evidence and target-specific pointwise reading combine contextual information with identifiable local trajectories.
- We develop OGRR, which combines a neural residual with a gated observation-reference correction to improve the initial reconstruction. It preserves visible measurements and can improve existing imputation backbones with their weights frozen, extending its use beyond the full framework.
- We evaluate the framework across six domains and four missingness configurations, obtaining best or second-best results in multiple settings. Paired component studies and shared-refinement comparisons examine the contributions of patch evidence and observation-guided correction, including OGRR's independent recovery capability.

2. Related Work

Multivariate time-series imputation. Deep imputation models exploit temporal dependencies and cross-variable relationships through different reconstruction mechanisms. BRITS uses bidirectional recurrent estimation (Cao et al., 2018), and SAITS employs diagonally masked self-attention (Du et al., 2023). CSDI models conditional distributions with score-based diffusion (Tashiro et al., 2021), while ImputeFormer introduces low-rank spatiotemporal attention (Nie et al., 2024). Our framework integrates contextual estimation with explicit patch evidence and a subsequent observation-guided correction stage.

Temporal and cross-variable representation. PatchTST represents local trajectories using temporal patches (Nie et al., 2023); iTransformer models dependencies between variate tokens (Liu et al., 2024a). T1 binds temporal feature channels to variable-attention heads (Park et al., 2026). PCE adopts this channel–head geometry as its variable-signature attention building block. MPEC and PEGR extend contextual reconstruction with source-organized fragments and pointwise reading, while OGRR revisits the resulting numerical estimate using visible observations.

Structured temporal modeling. TimeMixer++ combines multiple representation and mixing operations to capture temporal patterns and analyzes its components through ablation, representation, efficiency and sensitivity studies (Wang et al., 2025). For imputation, we organize information around point context, patch evidence and observation-guided correction. Capacity-matched controls, shared refinement and mask-defined error strata connect these design choices to measured reconstruction behavior.

3. PRISM

3.1. Problem Formulation

Given a time-series batch $\mathbf{X} \in \mathbb{R}^{B\times L\times C}$ with batch size B, window length L and C variables, let $\mathbf{M} \in \{0,1\}^{B\times L\times C}$ denote its observation mask. An entry $m_{n,t,c} = 1$ indicates an observed value of variable c at time t in sample n; zero indicates a value to reconstruct. The input $\mathbf{X}_{\text{obs}}$ retains visible values and sets unavailable entries to zero. Our objective is to reconstruct the complete window from these values and their mask:

$$\widehat{\mathbf{X}} = \text{PRISM}(\mathbf{X}_{\text{obs}}, \mathbf{M}). \tag{1}$$

The output $\widehat{\mathbf{X}} \in \mathbb{R}^{B\times L\times C}$ preserves original measurements at observed entries and estimates missing entries.

As shown in Fig. 2, PRISM combines Point-Level Context Encoding (PCE), Multi-Source Patch Evidence Construction (MPEC) and Pointwise Evidence-Guided Reconstruction (PEGR) to form an initial reconstruction $\mathbf{X}^{\text{i}}$. Observation-Guided Residual Refinement (OGRR) then corrects this estimate using visible measurements. Point context describes current-window dynamics, patch evidence supplies local trends, and observation guidance further corrects reconstructed values.

3.2. Point-Level Context Encoding

iTransformer organizes cross-variable interaction around variable trajectories (Liu et al., 2024a), while T1 transfers temporal patterns through a temporal CNN and channel–head binding (Park et al., 2026). Drawing on these ideas,

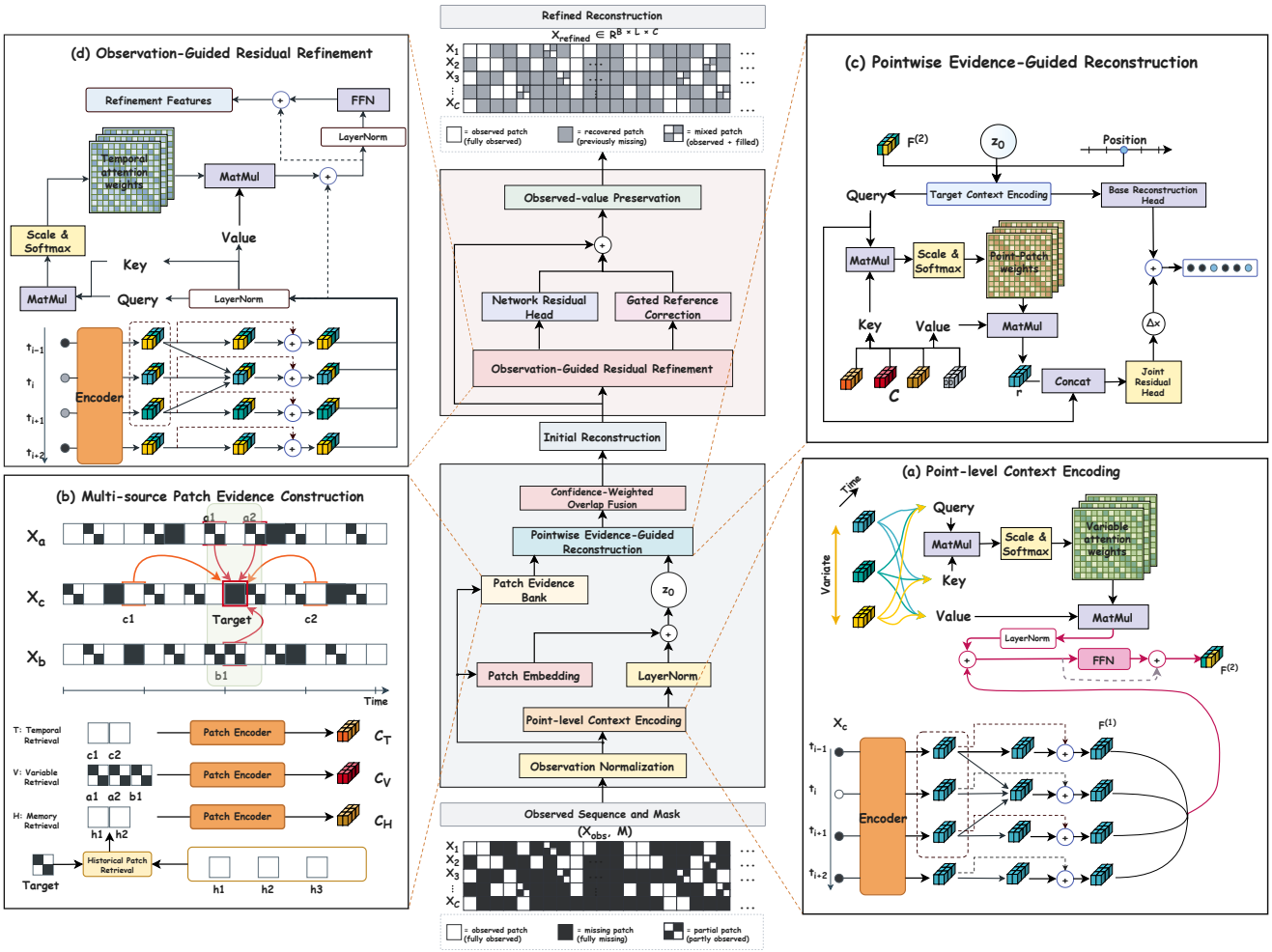


Figure 2. The framework of PRISM. The observation-normalized input first passes through Point-Level Context Encoding (a), which combines residual temporal convolution and cross-variable attention. Multi-Source Patch Evidence Construction (b) organizes temporal, cross-variable and eligible training-history fragments into a common evidence bank. Pointwise Evidence-Guided Reconstruction (c) reads this bank with a target-specific query and jointly uses the query and readout to predict a residual over the base estimate. Confidence-weighted overlap fusion forms the initial reconstruction. Observation-Guided Residual Refinement (d) then combines this estimate with visible observations and mask information through residual temporal convolution and temporal attention. A neural residual head and a gated observation-reference correction jointly refine the missing values, while the final mask operation preserves observed measurements.

PCE retains point-level temporal resolution and uses local dynamics together with variable relationships for both base reconstruction and subsequent patch-evidence reading. We call its temporal encoder Mask-Aware Temporal Convolution (MATC). Visible values, masks and visible-position relationships allow its updates to account for numerical variation and observation support. With input embedding H, encoding is

$$\mathbf{F}^{(1)} = H + \mathrm{MATC}(H),\ \mathbf{F}^{(2)} = \mathbf{F}^{(1)} + \mathrm{TSVA}(\mathbf{F}^{(1)}). \quad (2)$$

The tensor $\mathbf{F}^{(1)}$ represents temporal context and $\mathbf{F}^{(2)}$ represents integrated point context; both have dimensions $B \times L \times C \times d$. MATC produces temporal-convolution residuals; Temporal-Signature Variable Attention (TSVA) computes variable relationships from temporal feature signatures and produces cross-variable residuals that include feed-forward updates. Each feature channel supplies an attention pathway. Patch aggregation of this point context, combined with visible-value and mask embeddings, yields z_0, allowing point dynamics and local-fragment evidence to cooperate during reconstruction.

3.3. Multi-Source Patch Evidence Construction

PatchTST preserves local semantics through consecutive patches (Nie et al., 2023), while S4M supplements incomplete inputs with historical prototype representations (Peng et al., 2025). MPEC retains individual fragments as point-

queryable evidence and distinguishes temporal (T), cross-variable (V) and historical (H) sources. The target-patch index i contains its sample, temporal position and variable; c_i denotes its variable.

Temporal evidence T. Other patches of the same variable are organized into directional distance bands; each band scores and selects one fragment:

$$s_{\mathrm{T}}(i,k) = \begin{cases} -\operatorname{dist}(i,k), & k \in \mathcal{B}_{\mathrm{near}}(i), \\ \langle \phi_{\mathrm{T}}(i), \psi_{\mathrm{T}}(k) \rangle / \sqrt{d}, & \text{otherwise}, \end{cases} \tag{3}$$
$$\mathcal{C}_{\mathrm{T}}(i) = \left\{ \mathrm{Top1}_{s_{\mathrm{T}}}(\mathcal{B}_{\mathrm{T,d}}(i)) \right\}_{d \in \mathcal{D}},$$

where $\mathcal{B}_{\mathrm{near}}(i)$ is the nearest band and dist measures patch distance. The mappings $\phi_{\mathrm{T}}, \psi_{\mathrm{T}}$ project target and source features to dimension d. The set $\mathcal{D}$ contains nonempty bands, $\mathcal{B}_{\mathrm{T,d}}(i)$ retains eligible patches with visible support, and Top1 selects the highest score. Near bands favor nearby structure; farther bands learn to match relevant dynamics.

Cross-variable evidence V. Variable relationships and current visibility jointly determine source scores and candidates:

$$s_{\mathrm{V}}(i,v) = a(c_i, v) + \tfrac{1}{2}\rho(i,v) + \tfrac{1}{2}\chi(i,v), \tag{4}$$
$$\mathcal{C}_{\mathrm{V}}(i) = \{ x_{v,\delta} : v \in \mathcal{S}_{\mathrm{V}}(i),\ \delta \in \{-1, 0, 1\} \},$$

where a is a learned variable relationship, ρ measures visibility across neighboring source fragments, and χ measures consistency at commonly visible positions of aligned patches. The source variables are $\mathcal{S}_{\mathrm{V}}(i) = \mathrm{TopK}_{v \neq c_i} s_{\mathrm{V}}(i,v)$; $x_{v,\delta}$ is a patch from source v at offset δ relative to the target. Only in-range fragments with visible support are retained.

Historical evidence H. Retrieval first locates an eligible same-variable historical group by key similarity, then scores and selects members within it:

$$s_{\mathrm{H}}(i,h) = \cos(e_i, e_h), \tag{5}$$
$$\mathcal{C}_{\mathrm{H}}(i) = \operatorname*{TopK}_{x_h \in \mathcal{A}_{\mathrm{H}}(i)} s_{\mathrm{H}}(i,h),$$

where e_i, e_h are mask-aware target and member keys, and $\mathcal{A}_{\mathrm{H}}(i)$ contains patches satisfying temporal, provenance and similarity requirements in the selected group. Queries and members use consistent coordinates, and individual members retain local numerical structure. The bank is built exclusively from training data. Candidates from all three sources form $\mathcal{C}$, carrying source identities, quality, visible support and eligibility flags.

3.4. Pointwise Evidence-Guided Reconstruction

PEGR lets each target point read patch evidence independently through three steps: conditional projection, evidence residuals and base reconstruction.

Conditional projection. As shown in Fig. 2(c), z_0, point context $\mathbf{F}^{(2)}$ and position embedding $\mathbf{P}$ jointly construct the query, while the candidate bank $\mathcal{C}$ is projected to keys and values in parallel:

$$q = \mathrm{Query}(z_0, \mathbf{F}^{(2)}, \mathbf{P}), \quad [K, V] = \mathrm{KeyValue}(\mathcal{C}), \tag{6}$$

where z_0 supplies the target patch's overall state, $\mathbf{F}^{(2)}$ retains temporal and variable context at the current point, and $\mathbf{P}$ distinguishes within-patch positions. The mapping KeyValue produces matching representations K and readable content V from the same candidate input.

Evidence reading and residuals. Point-to-Patch Evidence Attention (PPEA) reads the candidates, and the readout jointly determines the residual with the query:

$$r = \mathrm{PPEA}(q, K, V), \quad \Delta x = \mathrm{MLP}(\mathrm{LN}([q; r])), \tag{7}$$

where PPEA computes scaled dot-product scores between q and K, incorporates source priors, quality and eligibility, and uses Softmax-normalized weights to read V (Vaswani et al., 2017). The tensor r is the readout. Concatenated features $[q; r]$ pass through layer normalization LN and the MLP's nonlinear mapping to generate Δx, together implementing the diagram's Joint Residual Head. Candidate corrections are zero when no candidate is eligible.

Base reconstruction and fusion. In parallel with evidence reading, the Base Decoder produces a contextual estimate. Adding the residual gives patch-level reconstruction:

$$\mathbf{X}_{\mathrm{base}} = \mathrm{BaseDecoder}(z_0), \quad \mathbf{X}^{\mathrm{c}} = \mathbf{X}_{\mathrm{base}} + \Delta x, \tag{8}$$

where $\mathbf{X}_{\mathrm{base}}$ is the base estimate and $\mathbf{X}^{\mathrm{c}}$ contains patch-level values corrected by evidence. Base decoding and residual outputs include their respective scale conversions, so both are added in the same coordinates. Overlapping patches are then assembled into a complete window:

$$\mathbf{X}^{\mathrm{i}} = \mathrm{Fusion}(\mathbf{X}^{\mathrm{c}}), \tag{9}$$

where Fusion uses normalized confidence weights to combine patch estimates of the same position, excludes padding and preserves visible measurements. Thus $\mathbf{X}^{\mathrm{c}}$ becomes the complete initial reconstruction $\mathbf{X}^{\mathrm{i}} \in \mathbb{R}^{B \times L \times C}$ supplied to OGRR.

3.5. Observation-Guided Residual Refinement

Initial reconstruction integrates point context and patch evidence, yet its numerical values can deviate in local level, variation amplitude or regions near visible boundaries. OGRR targets these deviations through residual correction. It combines temporal context in the reconstructed trajectory with numerical anchors from original visible measurements, forming the neural and gated-reference correction paths in Fig. 2(d).

Observation-conditioned features. The reference $\mathbf{R}$ is derived from visible neighbors of the same variable: linear interpolation with two-sided observations, or the available value with one-sided support. Support U decays with observation distance and is zero when no reference exists. The discrepancy $D = U \odot (\mathbf{R} - \mathbf{X}^{\mathrm{i}})$ represents reference differences with observation support. We call this stage's local CNN Observation-Conditioned Temporal Convolution (OCTC). It jointly projects the initial estimate, visible values, mask, discrepancy and support, then extracts observation-conditioned features through local convolutional residuals:

$$\mathbf{H}_{\mathrm{ref}} = \mathrm{TRA}\big(\mathrm{OCTC}([\mathbf{X}^{\mathrm{i}}; \mathbf{X}_{\mathrm{obs}}; \mathbf{M}; D; U])\big)\,. \tag{10}$$

The tensor $\mathbf{H}_{\mathrm{ref}} \in \mathbb{R}^{B\times L\times d_r}$ corresponds to Refinement Features in the diagram. OCTC extracts neighboring changes conditioned on observations; Temporal Refinement Attention (TRA) relates positions across the window. Its temporal-attention and feed-forward updates each include normalization and residual connections (Vaswani et al., 2017). Thus $\mathbf{H}_{\mathrm{ref}}$ describes the initial trajectory, observation locations and available reference discrepancies.

Neural residual correction. The Network Residual Head predicts corrections at missing entries, with corrections at visible entries set to zero:

$$\Delta_{\mathrm{net}} = 2\tanh(\mathrm{MLP}_{\mathrm{net}}(\mathbf{H}_{\mathrm{ref}})/2)\,. \tag{11}$$

The mapping $\mathrm{MLP}_{\mathrm{net}}$ projects the state to the variables, and the hyperbolic tangent bounds corrections to $[-2, 2]$. This path learns reconstruction errors from temporal and cross-variable context without forcing corrections toward interpolation references. It is disabled for wholly unobserved windows.

Gated reference correction. The Gated Reference Correction adjusts reference discrepancies at missing entries with visible support, and is zero elsewhere:

$$\Delta_{\mathrm{ref}} = G(\mathbf{H}_{\mathrm{ref}}) \odot (\mathbf{R} - \mathbf{X}^{\mathrm{i}}). \tag{12}$$

The gate G is generated from $\mathbf{H}_{\mathrm{ref}}$ by a learned projection and hyperbolic tangent, taking values in $[-1, 1]$. Positive gates follow the reference discrepancy, while negative gates allow reverse adjustments. The trajectory and observation support jointly determine the correction strength. This path is disabled when the variable has no visible reference. Encoding U and D allows the gate to account for reference distance and disagreement with the current estimate.

Observed-value preservation and output. Both corrections are added to the initial reconstruction to produce the final output:

$$\widehat{\mathbf{X}} = \mathbf{X}^{\mathrm{i}} + \Delta_{\mathrm{net}} + \Delta_{\mathrm{ref}}. \tag{13}$$

Corrections are zero at visible entries to preserve original measurements, corresponding to Observed-Value Preservation in the diagram. Both output heads are initialized to zero so OGRR learns corrections from the initial estimate. Its input interface also accepts initial results from other imputation backbones.

3.6. Loss Functions

Let $\mathbf{E}$ denote the supervision mask. Initial reconstruction minimizes masked MSE with retrieval-key supervision and a numerical-level constraint; refinement minimizes weighted absolute error:

$$\mathcal{L}_{\mathrm{init}} = \mathrm{MSE}_{\mathbf{E}}(\mathbf{X}^{\mathrm{i}}, \mathbf{X}) + \lambda_k \mathcal{L}_{\mathrm{key}} + \lambda_l \mathcal{L}_{\mathrm{level}}. \tag{14}$$

$$\mathcal{L}_{\mathrm{refine}} = \frac{\|w \odot \mathbf{E} \odot (\widehat{\mathbf{X}} - \mathbf{X})\|_1}{\|\mathbf{E}\|_1}. \tag{15}$$

The term $\mathcal{L}_{\mathrm{key}}$ combines retrieval-key alignment and contrastive supervision; $\mathcal{L}_{\mathrm{level}}$ constrains numerical levels. We set $\lambda_k = 0.1$ and $\lambda_l = 1$, and w denotes the loss weights combining the implemented normalization rules. The initial reconstruction network, historical bank and retrieval encoders remain frozen during OGRR training.

4. Experiments

4.1. Datasets and Evaluation Protocol

We evaluate PRISM on ETTh1, Weather, Exchange-Rate, BeijingAirQuality, PhysioNet2012 and PAMAP2, covering industrial, meteorological, financial, environmental, clinical and wearable-sensor measurements. The prepared data use 7,21,8,12,35 and 40 variables, respectively. BeijingAirQuality treats the twelve stations as related channels. Native observation masks are retained; patient, subject and record boundaries are respected when constructing windows. All statistics used for scaling and history construction come from training data.

Artificial missingness is controlled by p_s and b. The parameter p_s controls the number of missing-event starts, while b specifies how many consecutive time points each event hides. For each variable over a timeline of length L, we uniformly sample $\mathrm{round}(p_s L)$ distinct starts from positions that can accommodate a complete missing block. An event starting at t hides the b consecutive values from t to $t + b - 1$. Blocks from different starts may overlap, so the realized missing fraction depends jointly on p_s, b and block overlap. For example, $p_s = 10\%$ and $b = 8$ sample a number of starts equal to 10% of the timeline length and hide eight consecutive values per start. We evaluate four configurations: $10\%/8$, $15\%/8$, $20\%/8$ and $10\%/16$. Averaged across the six datasets and twenty test masks, these configurations produce approximately 57%, 73%, 83% and 81% artificial missingness, respectively, excluding native missingness.

Dataset	Block masking	PRISM (Ours)		BRITS (2018)		CSDI (2021)		ImputeFormer (2024)		TimesNet (2023)		T1 (2026)	
	p_s / b	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
ETTh1	10% / 8	0.1168	0.2055	0.5188	0.4990	0.1872	0.2473	0.2164	0.2914	0.3029	0.3517	0.1323	0.2278
	15% / 8	0.1722	0.2565	0.6529	0.5788	0.3038	0.3277	0.3057	0.3512	0.4208	0.4121	0.1936	0.2805
	20% / 8	0.2619	0.3227	0.7771	0.6397	0.3834	0.3945	0.4797	0.4410	0.5825	0.4817	0.3125	0.3520
	10% / 16	0.3350	0.3720	0.8855	0.6801	0.4517	0.4291	0.5322	0.5002	0.6397	0.5124	0.3895	0.3987
Weather	10% / 8	0.0424	0.0435	0.0616	0.0685	0.0427	0.0361	0.0550	0.0547	0.0670	0.1062	0.0474	0.0646
	15% / 8	0.0525	0.0539	0.0719	0.0843	0.0506	0.0435	0.0636	0.0641	0.0829	0.1226	0.0608	0.0775
	20% / 8	0.0646	0.0703	0.0911	0.1079	0.0665	0.0589	0.0878	0.0808	0.1090	0.1466	0.0835	0.1041
	10% / 16	0.0786	0.0929	0.1057	0.1255	0.0776	0.0731	0.0919	0.0967	0.1621	0.1950	0.1237	0.1488
Exchange	10% / 8	0.0092	0.0578	1.2295	0.8475	0.0098	0.0575	0.0576	0.1501	0.0196	0.0918	0.0084	0.0515
	15% / 8	0.0154	0.0739	1.2301	0.8567	0.0169	0.0752	0.0945	0.1931	0.0241	0.1006	0.0140	0.0662
	20% / 8	0.0542	0.0985	1.1269	0.8653	0.0447	0.1056	0.1195	0.2175	0.0577	0.1234	0.0476	0.0964
	10% / 16	0.1623	0.1720	1.2345	0.9078	0.1577	0.1864	0.2713	0.3147	0.1997	0.2001	0.1636	0.1774
Beijing AQ	10% / 8	0.0772	0.1446	0.1023	0.1684	0.0776	0.1419	0.0868	0.1502	0.2273	0.2927	0.0861	0.1585
	15% / 8	0.0948	0.1631	0.1185	0.1843	0.0915	0.1575	0.0998	0.1658	0.3430	0.3638	0.1285	0.1971
	20% / 8	0.1228	0.1882	0.1426	0.2074	0.1176	0.1814	0.1244	0.1877	0.5200	0.4459	0.2255	0.2643
	10% / 16	0.1468	0.2069	0.1541	0.2150	0.1359	0.1950	0.1533	0.2101	0.7528	0.5501	0.3924	0.3617
PhysioNet	10% / 8	0.3868	0.3434	0.5181	0.4190	0.5745	0.4547	0.3742	0.3337	0.5314	0.4412	0.4871	0.4144
	15% / 8	0.4765	0.4038	0.5882	0.4680	0.6721	0.5119	0.4614	0.3936	0.6574	0.5078	0.5970	0.4727
	20% / 8	0.5648	0.4553	0.6522	0.5090	1.0306	0.6609	0.5362	0.4440	0.7701	0.5579	0.7328	0.5394
	10% / 16	0.5818	0.4699	0.6863	0.5383	0.9781	0.6614	0.5494	0.4573	0.8291	0.5854	0.7736	0.5610
PAMAP2	10% / 8	0.2276	0.1816	0.6593	0.4508	0.4340	0.2374	0.4385	0.2525	0.5484	0.3305	0.2299	0.1867
	15% / 8	0.3267	0.2308	0.7439	0.4847	0.5172	0.2796	0.5474	0.2979	0.6128	0.3548	0.3367	0.2397
	20% / 8	0.4669	0.2927	0.8260	0.5248	0.6143	0.3225	0.6146	0.3350	0.7205	0.3896	0.5020	0.3066
	10% / 16	0.5320	0.3346	0.9201	0.5719	0.6497	0.3514	0.6713	0.3693	0.7834	0.4232	0.5913	0.3536

Table 1. Imputation results across six datasets and four missingness configurations. Each training run averages twenty shared test masks; PRISM reports the average over training seeds2023,2024 and 2025, while baselines retain their original training seeds. Best results are red; second-best results are blue and underlined.

For each dataset and missingness configuration, twenty fixed random seeds generate twenty distinct sets of missing positions using this procedure. All models are evaluated on the same twenty test masks for a fair comparison. Hash digests verify shared-mask identity, and the masks remain fixed across models and test outcomes. Native missingness is retained; MSE and MAE are computed only at originally observed values that are artificially hidden. Errors are measured in coordinates standardized using training-set statistics and averaged over the twenty test masks.

4.2. Main Results

Table 1 compares PRISM with five baselines across six datasets and four masking configurations. Each model has adjacent MSE and MAE columns; rows group the four scenarios within each dataset. The long-block and higher-start-rate cases test the model with reduced visible support.

References

Wei Cao, Dong Wang, Jian Li, Hao Zhou, Lei Li, and Yitan Li. Brits: Bidirectional recurrent imputation for time series, 2018. 1, 3

Zhengping Che, Sanjay Purushotham, Kyunghyun Cho, David Sontag, and Yan Liu. Recurrent neural networks for multivariate time series with missing values. *Scientific Reports*, 8:6085, 2018. 1

Andrea Cini, Ivan Marisca, and Cesare Alippi. Filling the g_ap_s: Multivariate time series imputation by graph neural networks. In *International Conference on Learning Representations*, 2022. 1

Wenjie Du, David Cote, and Yan Liu. Saits: Self-attention-based imputation for time series, 2023. 1, 3

Mingzhe Liu, Han Huang, Hao Feng, Leilei Sun, Bowen Du, and Yanjie Fu. PriSTI: A conditional diffusion framework for spatiotemporal imputation. In *IEEE International Conference on Data Engineering*, 2023. 1, 2

Yong Liu, Tengge Hu, Haoran Zhang, Haixu Wu, Shiyu Wang, Lintao Ma, and Mingsheng Long. itransformer: Inverted transformers are effective for time series forecasting, 2024a. 2, 3

Zijin Liu, Xiang Zhao, and You Song. RDPI: A refine diffusion probability generation method for spatiotemporal data imputation, 2024b. 2

Tong Nie, Guoyang Qin, Wei Ma, Yuewen Mei, and Jian Sun. Imputeformer: Low rankness-induced transformers for generalizable spatiotemporal imputation, 2024. 1, 3

Yuqi Nie, Nam H. Nguyen, Phanwadee Sinthong, and Jayant Kalagnanam. A time series is worth 64 words: Long-term forecasting with transformers, 2023. 2, 3, 4

Dongik Park, Hyunwoo Ryu, Suahn Bae, Keondo Park, and Hyung-Sin Kim. T1: One-to-one channel-head binding for multivariate time-series imputation, 2026. 1, 3

Jing Peng, Meiqi Yang, Qiong Zhang, and Xiaoxiao Li. S4M: S4 for multivariate time series forecasting with missing values. In *International Conference on Learning Representations*, 2025. 4

Attila Reiss and Didier Stricker. Introducing a new benchmarked

dataset for activity monitoring. In *IEEE International Symposium on Wearable Computers*, pages 108–109, 2012. 1

Ikaro Silva, George Moody, Daniel J. Scott, Leo A. Celi, and Roger G. Mark. Predicting in-hospital mortality of ICU patients: The PhysioNet/Computing in Cardiology Challenge 2012. In *Computing in Cardiology*, pages 245–248, 2012. 1

Yusuke Tashiro, Jiaming Song, Yang Song, and Stefano Ermon. Csdi: Conditional score-based diffusion models for probabilistic time series imputation, 2021. 2, 3

Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Lukasz Kaiser, and Illia Polosukhin. Attention is all you need, 2017. 5, 6

Shiyu Wang, Jiawei Li, Xiaoming Shi, Zhou Ye, Baichuan Mo, Wenze Lin, Shengtong Ju, Zhixuan Chu, and Ming Jin. Timemixer++: A general time series pattern machine for universal predictive analysis, 2025. 1, 2, 3

Haixu Wu, Tengge Hu, Yong Liu, Hang Zhou, Jianmin Wang, and Mingsheng Long. TimesNet: Temporal 2d-variation modeling for general time series analysis. In *International Conference on Learning Representations*, 2023. 1, 2